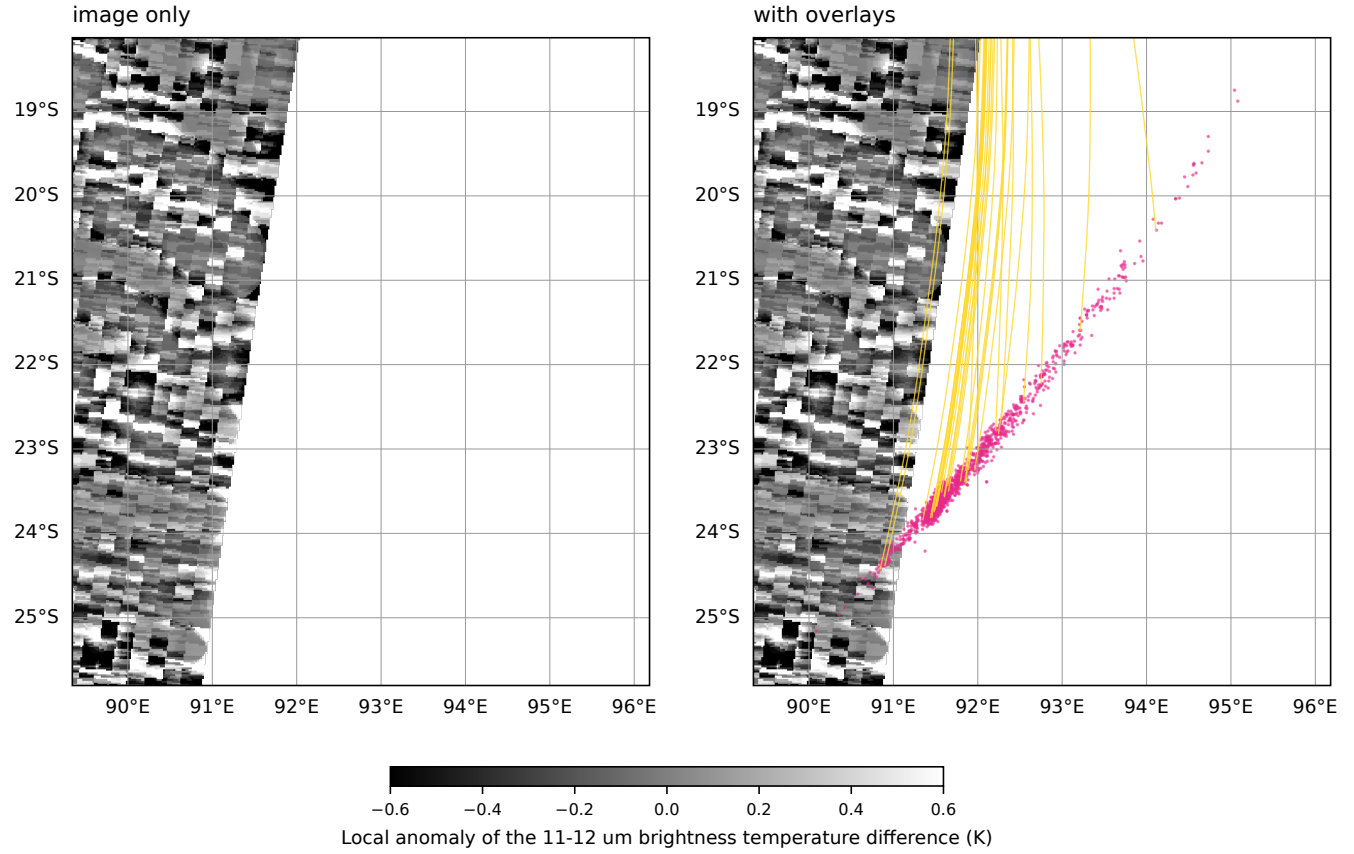
NOAA-18 AVHRR, 22:33 UTC 7 March 2014: aircraft probably just outside the archived swath, contrail not assessable



- likely aircraft position at the scan time (2,000 posterior samples)
- where a contrail up to 60 min old would lie (35 posterior routes, FL350 wind)

Plain note: coarse heat picture from NOAA-18 at 22:33 UTC. The aircraft was most likely just outside the east edge of the picture. Any contrail would be too narrow to see at this edge.

Technical note: AVHRR/3 channels 4 (10.8 um) minus 5 (12.0 um), FIDUCEO FCDR v1.00 EASY (GAC-based; CEDA), FIDUCEO_FCDR_L1C_AVHRR_N18ALL_20140307213614_20140307231816. P(aircraft in swath) = 0.678 with the nominal 1500 km half-swath, 0.0667 with the measured archived east edge (1468-1471 km); P(contrail <= 15/30/60 min old in swath) = 0.124/0.676/0.926 (measured edge). Scan angle 55.2 deg (view zenith 68.6 deg); GAC across-scan sample spacing at the edge 19-23 km; Sun -18.9 deg, Moon -76.2 deg. Posterior: run J, BTO+BFO case (waypoints 0.9982, free 0.0013, reproduced-radar 0.0003, descent-climb 0.0002; 148,000 routes, linear interpolation between 600 s vertices); radar-tracked flight 16:42:27-18:22:12 from the Geoscience Australia digital military radar track. Positions use the granule's own scan time at each position and a parallax shift for FL350 (away from the sub-satellite point by 10.67 km x tan(view zenith)). Contrail loci use no contrail spreading; drift = ERA5 wind at FL350 (238.4 hPa) or FL400 (187.5 hPa) x contrail age. Image: brightness temperature difference minus its 21-pixel running median, on a 0.02 deg (15 km search radius) nearest-neighbour grid. ERA5 (Schmidt-Appleman, efficiency 0.3): no contrail formation at FL300 or FL350 at this time; formation at FL400 but ice relative humidity < 100 %, so no persistence.